

**B.Tech. Degree I Semester Regular Examination November 2023****EE/EC/CS/IT 23-200-0103B INTRODUCTION TO ELECTRONICS DEVICES AND CIRCUITS**  
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

**Course Outcomes**

On successful completion of the course, the students will be able to:

CO1: To understand the working principle of various semiconductor devices.

CO2: To apply the acquired knowledge to the use of semiconductor devices in various applications.

CO3: To design simple electronic circuits for a given application.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

**PO – Programme Outcome****PART A**(Answer *ALL* questions)

|    |                                                                                                                                                      | (5 × 2 = 10) | Marks | BL | CO | PO |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|----|----|----|
| I. | (a) Write a short note on doping in semiconductors. What is the need for doping in semiconductors?                                                   |              | 2     | L1 | 1  | 1  |
|    | (b) Draw and explain the characteristics of a PN junction diode.                                                                                     |              | 2     | L1 | 1  | 1  |
|    | (c) Explain thermal runaway in transistors.                                                                                                          |              | 2     | L1 | 2  | 1  |
|    | (d) A three-stage amplifier has first, second and third stage voltage gains of 10, 15 and 20 respectively. Find the overall voltage gain in decibel. |              | 2     | L3 | 2  | 2  |
|    | (e) What are the benefits of feedback in amplifiers and oscillators?                                                                                 |              | 2     | L2 | 3  | 1  |

**PART B**

(4 × 10 = 40)

|     |                                                                              |    |    |   |   |
|-----|------------------------------------------------------------------------------|----|----|---|---|
| II. | Explain the working of a full wave bridge rectifier with suitable waveforms. | 10 | L1 | 1 | 2 |
|-----|------------------------------------------------------------------------------|----|----|---|---|

**OR**

|      |                                                                                       |   |    |   |   |
|------|---------------------------------------------------------------------------------------|---|----|---|---|
| III. | (a) How materials are classified based on their energy band diagram?                  | 6 | L2 | 1 | 2 |
|      | (b) Explain, with suitable waveforms, the working of a capacitor filter.              | 4 | L1 | 1 | 1 |
| IV.  | (a) Explain the various current components of a BJT. Mention the relation among them. | 6 | L1 | 2 | 1 |
|      | (b) Write a short note on how a transistor works as a switch.                         | 4 | L1 | 2 | 1 |

**OR**

|    |                                                                                               |    |    |   |   |
|----|-----------------------------------------------------------------------------------------------|----|----|---|---|
| V. | Draw and explain the input and output characteristics of BJT in Common Emitter configuration. | 10 | L1 | 2 | 1 |
|----|-----------------------------------------------------------------------------------------------|----|----|---|---|

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|           |                                                                                                               | Marks | BL | CO | PO |
|-----------|---------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| VI.       | Explain the circuit and working of a Common Emitter RC coupled amplifier and its frequency response behavior. | 10    | 1  | 2  | 1  |
| <b>OR</b> |                                                                                                               |       |    |    |    |
| VII. (a)  | Describe the working of a Common Source FET amplifier.                                                        | 6     | L1 | 2  | 1  |
| (b)       | Explain the four types of feedback amplifiers with block diagram.                                             | 4     | L2 | 2  | 1  |
| VIII.     | Explain the circuit and working of an RC phase shift oscillator.                                              | 10    | L1 | 3  | 1  |
| <b>OR</b> |                                                                                                               |       |    |    |    |
| IX.       | How an astable multivibrator works? Explain with a neat circuit diagram.                                      | 10    | L2 | 3  | 1  |

Bloom's Taxonomy Levels

L1 – 73.3%, L2 – 24.5%, L3 – 2.2%.

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